

INFO-F404 – Oral Exam

List of questions (January 2026)

Exam Guidelines

- The exam can be taken either in English or French.
- The exam is individual and oral. Use of the curriculum guide, notes, and slides is permitted.
- Two questions, from the list below, will be asked, each related to a different chapter. A few minutes will be provided to optionally consult the curriculum guide, notes, or slides (e.g., to locate a numerical example or recall the main idea of a proof) before answering orally.
- During the response, a blank sheet of paper may be used to write down explanations if needed. A calculator will also be available for use if required.
- After the response, a discussion will follow, which may include sub-questions on the same topic, further details, or examples.
- The goal is to evaluate the depth of understanding, ranging from superficial (fail) to very deep (success).
- It is not essential to replicate specific examples, arguments, or notations from the course. The focus will be on the correctness and comprehension of the responses.

Uniprocessor scheduling

- Q.1** Give a periodic implicit deadline systems not schedulable (using RM) with $U < 1$.
- Q.2** Prove the property: Let τ be a periodic task system, and let S_1 and S_2 be two schedules of work-conserving schedulers for τ . The schedule S_1 is idle at instant t iff the schedule S_2 is idle at instant t , for all $t \geq 0$. What is the purpose of this property?
- Q.3** Prove the optimality of RM in the FTP class.
- Q.4** For the FTP class, synchronous systems, arbitrary deadline: show the non-optimality of RM & DM schedulers. Exhibit the important phenomenon which exists for arbitrary schedules but not for implicit or constrained deadlines.
- Q.5** For RM, synchronous systems and constrained deadlines: provide the equation and the fixed-point iteration process for the computation of the response time of the **first** job of τ_i . Discuss the convergence of the process and suggest alternative initialization values.
- Q.6** For RM, synchronous systems and constrained deadlines: find a system with 3 tasks ($\tau_1 > \tau_2 > \tau_3$) where the equation

$$r_3^1 = C_3 + \sum_{j=1}^2 \left\lceil \frac{r_3^1}{T_j} \right\rceil \cdot C_j$$

has **two** (positive) solutions.

Q.7 Consider a synchronous constrained-deadline system with three tasks:

$$\tau_1 = (C_1 = 2, T_1 = 5, D_1 = 4), \tau_2 = (C_2 = 3, T_2 = 8, D_2 = 7), \tau_3 = (C_3 = 1, T_3 = 10, D_3 = 9).$$

Calculate the response time of the first job of τ_3 under RM and verify schedulability.

Q.8 For the FTP class, synchronous systems, arbitrary deadline: provide and prove correct a feasibility interval.

Q.9 For deterministic schedulers and discrete time space, prove that feasible schedules of periodic tasks are periodic.

Q.10 For the FTP class, asynchronous systems, constrained deadline: prove correct the feasibility interval $[0, S_n + P)$. Explain the meaning of the S_i instants.

Q.11 For the FTP class ($\tau_1 > \tau_2 > \tau_3$), asynchronous systems, constrained deadline: compute the S_i values for the following system (O, C, T) :

$$\tau_1 : (10, 2, 10), \quad \tau_2 : (0, 1, 5), \quad \tau_3 : (1, 1, 3)$$

and then check the schedulability.

Q.12 For the FTP class, asynchronous systems, constrained deadline: provide and prove correct the main property foundation of the optimal AUDSLEY's priority assignment.

Q.13 Consider the following task set:

$$(O_1 = 3, C_1 = 2, D_1 = 2, T_1 = 4; \quad O_2 = 0, C_2 = 3, D_2 = 4, T_2 = 8)$$

In the framework of the FTP class (and asynchronous systems, constrained deadline), determine if τ_1 is lowest-priority viable.

Q.14 For a synchronous implicit-deadline task system, demonstrate the *strong* optimality of EDF by constructing a counterexample where RM fails but EDF succeeds.

Q.15 Prove the optimality of EDF.

Q.16 For EDF, implicit deadline systems: prove the fact that $U \leq 1$ is necessary and sufficient.

Q.17 For EDF, synchronous systems, constrained deadline: provide the feasibility interval and the procedure to determine its upper bound. Provide a formal proof of why this interval is sufficient to guarantee the schedulability of the system.

Q.18 LLF: define the notion of laxity, the priority management, and the thrashing situation using an example.

Q.19 Build an LLF schedule for the following system (C, T, D) : $(2, 6, 4)(4, 5, 6)$.

Q.20 For what type(s) of tasks and which scheduling algorithm(s) is the response time of the first job of each task sufficient to conclude the schedulability of the task?

Q.21 Provide at least three different feasibility intervals, specifying the type of tasks and the scheduler involved for each.

Multiprocessor scheduling

- Q.22** For global scheduling, prove that no online optimal scheduler exists.
- Q.23** Explain why the task-partitioning problem for implicit-deadline periodic tasks on identical processors can be interpreted as a special case of the bin-packing problem. For $m = 2$, provide a small task set for which First-Fit Decreasing Utilisation (FFDU) and Worst-Fit Decreasing Utilisation (WFDU) produce different partitions, and discuss which of the two heuristics yields the better assignment with your example, where 'better' is strictly defined as the ability to achieve a feasible assignment using *fewer* processors for the same task set.
- Q.24** Define the Next-Fit Decreasing Utilisation (NFDU) heuristic in the context of partitioned EDF scheduling. Apply NFDU to the following task set: $(C, T) = (5, 10), (2, 5), (1, 4), (1, 4), (1, 8), (1, 8), (1, 8), (1, 8)$. Show the resulting partition and comment on why NFDU may be highly sub-optimal, even when every task has utilisation at most 0.5.
- Q.25** Define the concept of scheduling anomalies, illustrate the phenomenon for global FTP scheduling and the period parameter.
- Q.26** In the context of PFair scheduling, define lag and explain the requirement $-1 < \text{lag}(\tau, t) < 1$. Show, on a small example with two periodic tasks, that enforcing this bound ensures that each job completes within its deadline, and explain intuitively why PFair scheduling is optimal for periodic implicit-deadline systems.
- Q.27** Given the task set $(C, T) = (1, 3), (2, 6), (5, 15), (6, 18)$, construct the global DM schedule on two processors over interval $[0, 18]$. Indicate all preemptions and migrations. State whether any deadline is missed and justify your conclusion.
- Q.28** Provide the basic sufficient condition for EDF schedulability of implicit deadline tasks, present the $\text{EDF}^{(k)}$ improvement.
- Q.29** Consider the sporadic implicit-deadline system $(C, T) = (7, 10), (3, 12), (3, 12), (2, 8), (1, 5)$. Sort the tasks by decreasing utilisation. For each value $k \in \{1, \dots, 5\}$, compute the required number of processors according to the $\text{EDF}^{(k)}$. Identify the optimal value of k and comment on the improvement compared to pure global EDF.
- Q.30** Build a PFair schedule for the following system upon two processors $(C, T): (2, 3)(2, 3)(4, 6)$. Indicate all preemptions and migrations.
- Q.31** Build a global EDF schedule for the following system upon two processors $(C, T): (1, 2)(3, 6)(6, 12)$. Indicate all preemptions and migrations.
- Q.32** Provide a necessary and sufficient condition for the feasibility of implicit deadline systems upon multiprocessors. Provide a rigorous formal proof for both the necessity and the sufficiency of this condition.
- Q.33** To demonstrate that partitioning and global scheduling are incomparable, provide a specific task set and a configuration of m processors such that a feasible partition exists, yet no global Fixed Job Priority (FJP) scheduler can successfully schedule the

set. Rigorously prove that **every** possible global FJP priority assignment leads to a deadline miss for this task set.

Parallel algorithms

Q.34 Provide **Amdahl's** law and illustrate its consequences.

Q.35 Bitonic Mergesort: principles, algorithms, complexity.

Q.36 Please apply the bitonic pre-processing process in order to convert this **arbitrary sequence** (length $n = 2^3$): $\langle 8, 7, 9, 5, 10, 3, 12, 1 \rangle$ into a bitonic sequence of length 8.

Concurrency

Q.37 Solution to the producer-consumer problem, bounded buffers.

Q.38 Describe the Bakery algorithm.

Q.39 Two questions related to the Bakery algorithm:

- Construct a scenario showing that the ticket numbers can be **unbounded** or provide an argument showing this is not possible.
- Construct a scenario showing that the ticket numbers can be **identical** or provide an argument showing this is not possible.

Q.40 Describe the **Ricart-Agrawala** algorithm (distributed critical sections).

Q.41 Starvation definition. Illustrate the phenomenon using one of **Dekker's** attempts.

Q.42 Live-lock definition. Illustrate the phenomenon using one of **Dekker's** attempts.

Q.43 Describe the Byzantine Generals algorithm.